

CPEN 416 / EECE 5710 Lecture 10

Quantum compilation I

Tuesday 14 October 2025

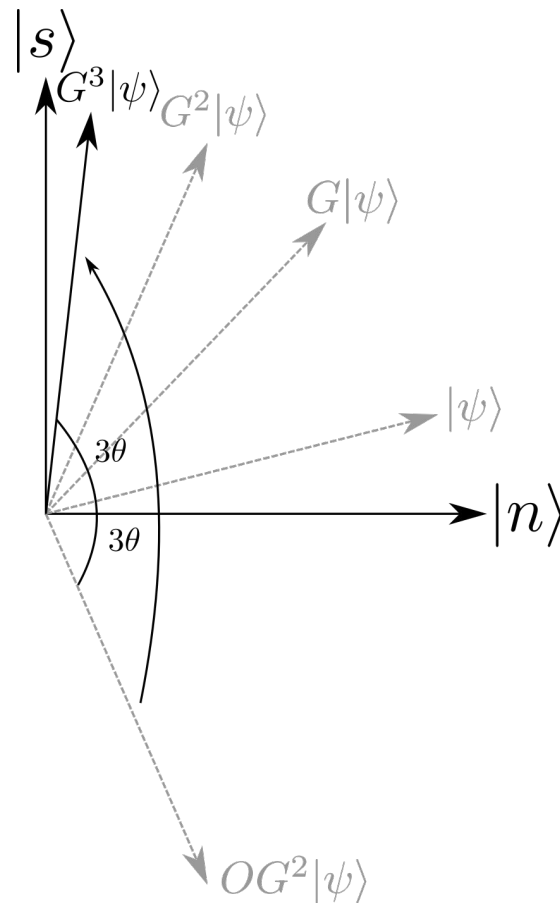
Announcements

- Quiz 4 today
- Final project group + paper selection **due 17:00 Thursday**

Last time

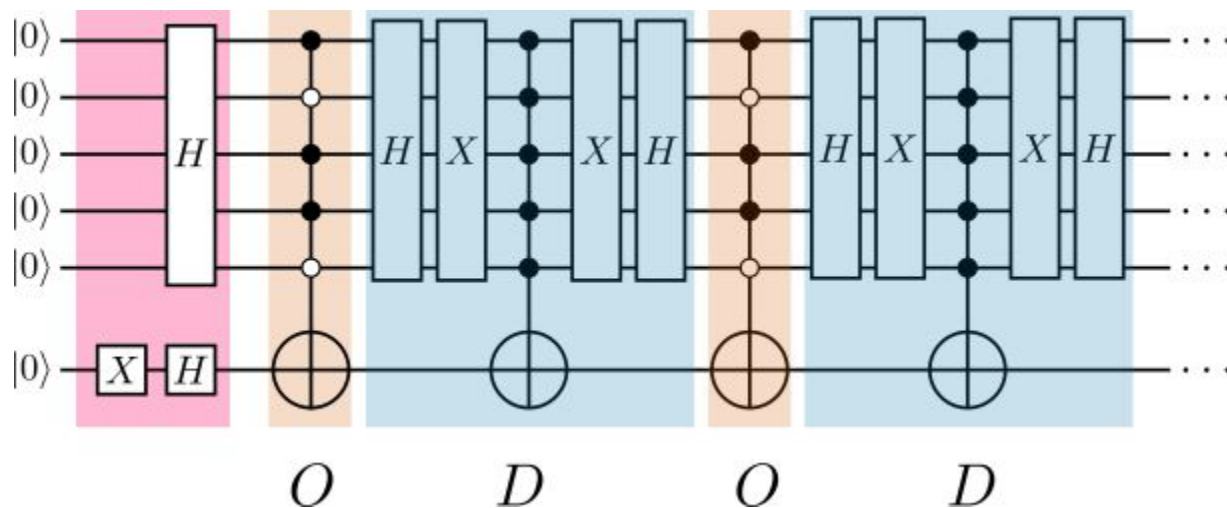
We discussed how Grover search of an unstructured space has a square-root speedup in **query complexity** over classical methods.

$$O(2^n) \rightarrow O(\sqrt{2^n})$$



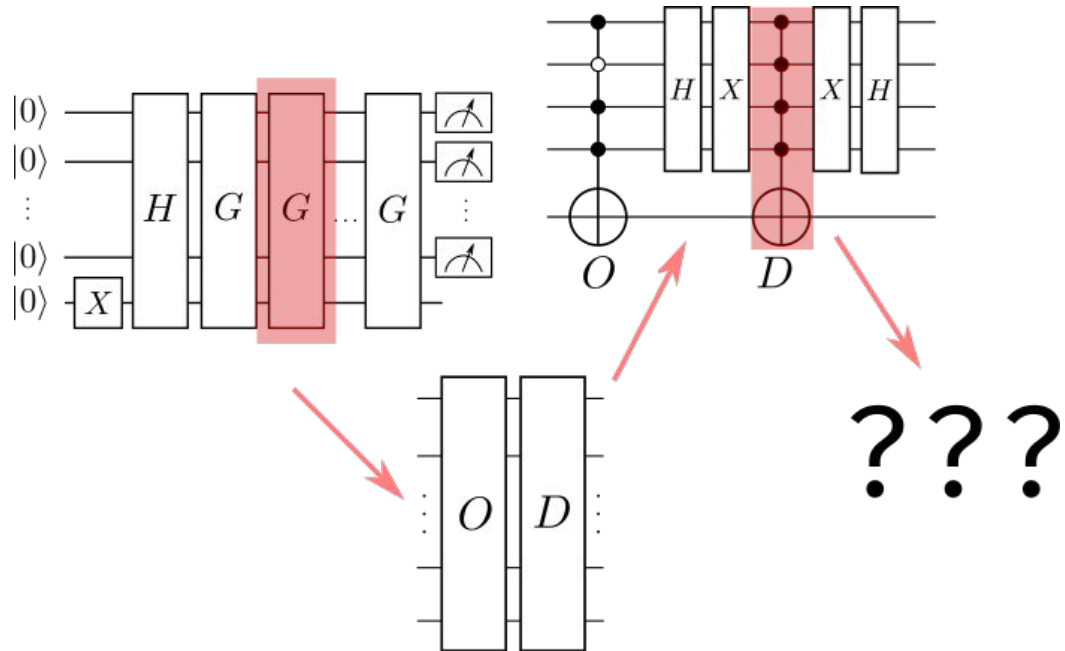
Last time

We implemented Grover's algorithm in PennyLane



Last time

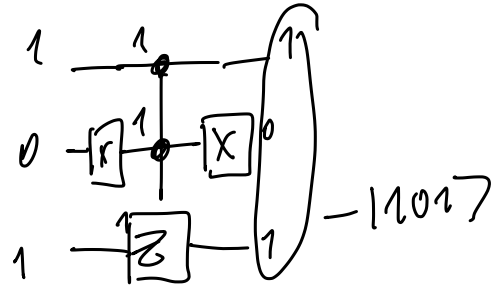
We finished off with this picture



Quiz

vazirani

$$CCZ |111\rangle = -|111\rangle$$



$$\vec{s} = 011 \quad f(\vec{s}) = 1$$

$$|011\rangle|0\rangle \rightarrow |011\rangle|1\rangle$$

$$U_f |\vec{x}\rangle|y\rangle = |\vec{x}\rangle|y \oplus f(\vec{x})\rangle$$

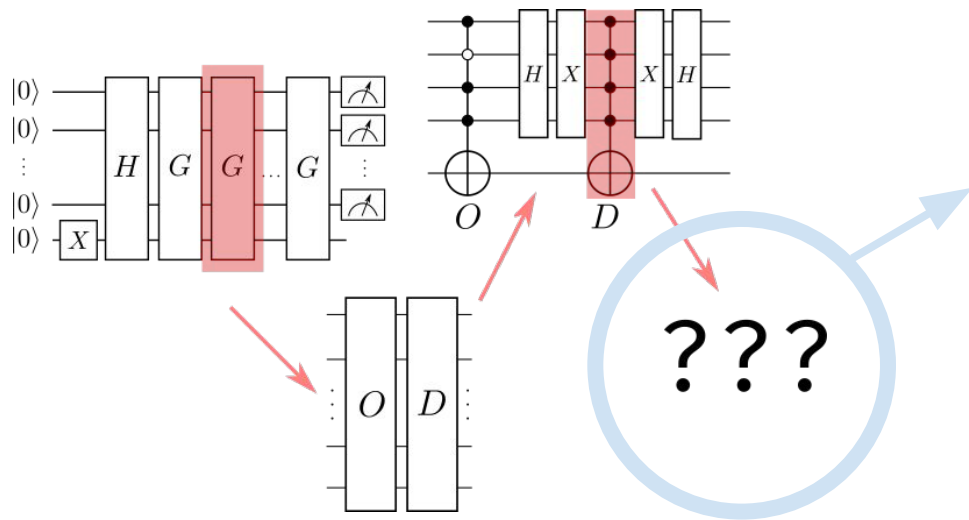
Today

Learning outcomes:

- define and calculate **cost metrics** of quantum circuits
- define and list key **universal** gate sets
- **decompose** selected single- and multi-qubit operations using one- and two-qubit universal gate sets

Thursday will involve hands-on component that builds on today's content (compiling Grover's algorithm).

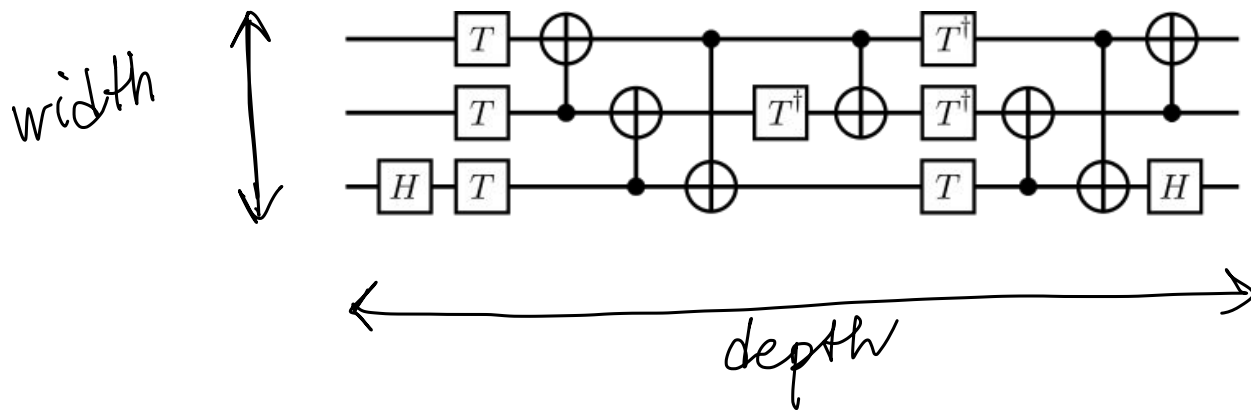
Quantum circuit synthesis and optimization



- How do we decide what goes here?
- What criteria determine whether our choice is “good”?
- How do we improve on our choices?

L01: cost metrics

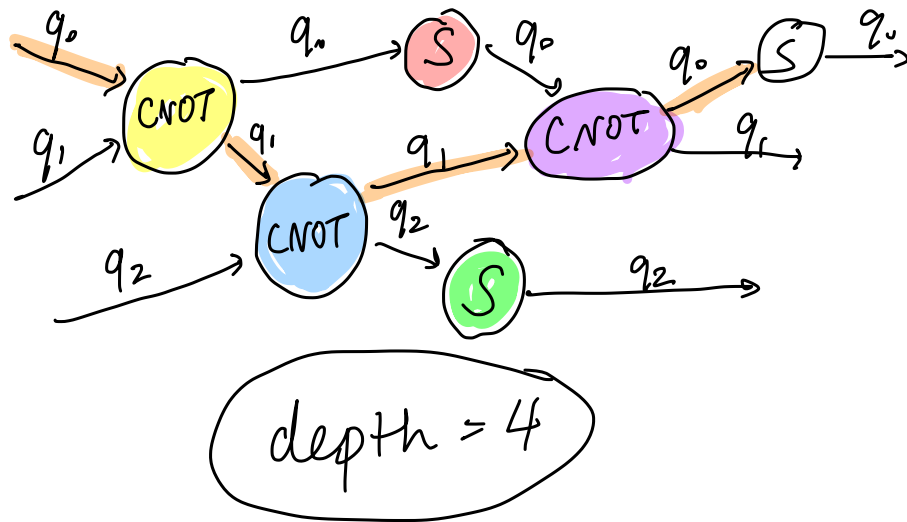
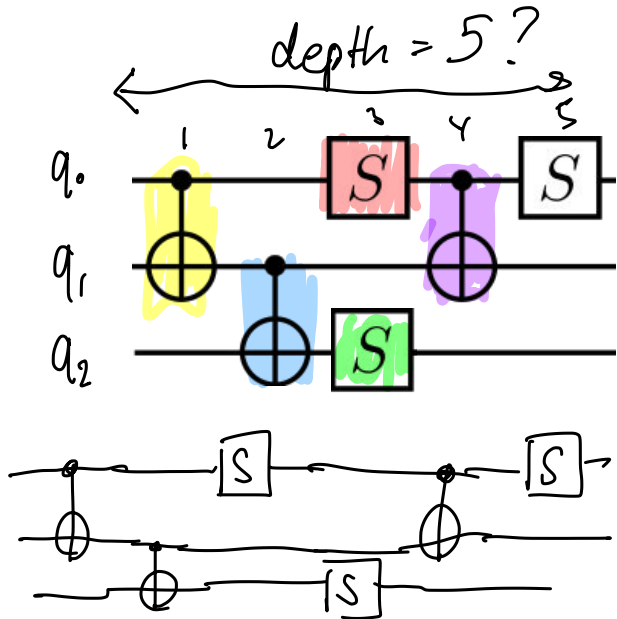
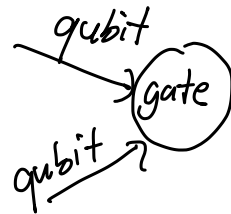
Width, depth, gate count



gate count: # 1 qubit gates # $T/T^\dagger$ count
 # 2 qubit gates # CNOT gates
 $\vdots$

Circuit depth

The circuit depth is the length of the longest path in its directed acyclic graph (DAG) representation.

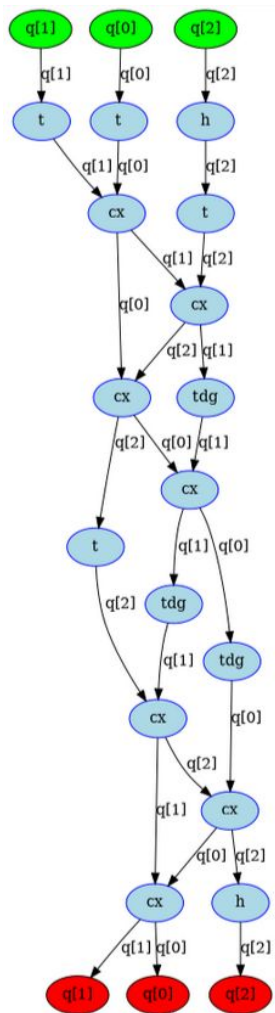
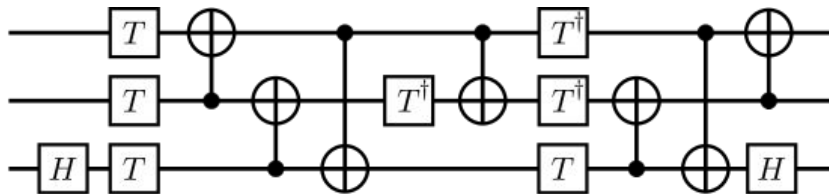


L01: cost metrics

Circuit depth

Generally don't do by hand; NetworkX
is your best friend.

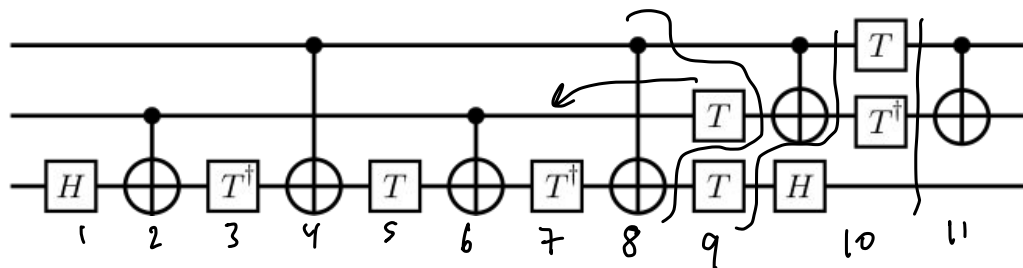
<https://networkx.org/>



(DAG generated
in Qiskit)

Exercise

Determine resource counts of the following circuit.

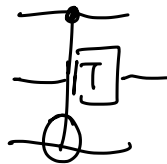


width? 3

depth? 11

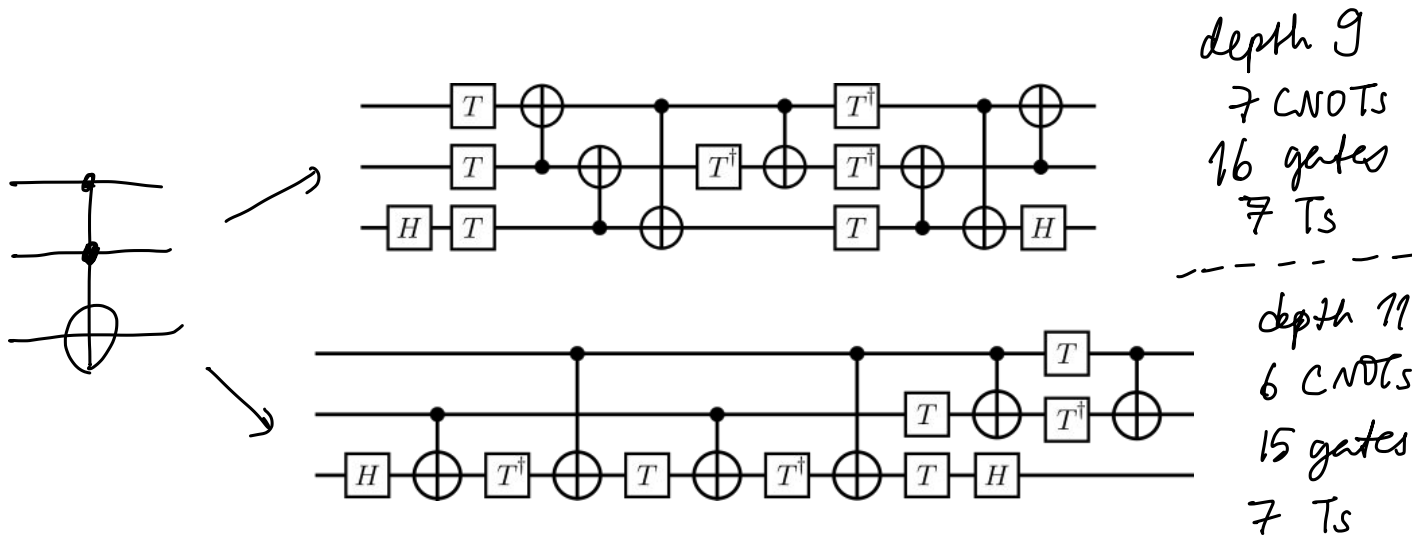
gate counts:

6 CNOTs
7 Ts
2 Hadamards
15



Circuit synthesis and decomposition

What do these circuits do?



Circuit synthesis

Given:

- arbitrary unitary matrix U
- a finite set of gates $\{G_k\}$
- a target precision, ϵ

Find

$$U' = C_1 C_2 \cdots C_m$$

s.t.

$$\|U - U'\| < \epsilon, C_i \in \{G_k\}$$

Universal gate sets

A $\{G_k\}$ for which this can be done is called a **universal gate set**.

Examples for a single qubit:

- Any two Pauli rotations (exact)
- Hadamard and T (approximate)

$H T T T H T H T T T H T \dots$

Theoretical bound: $O(\log^c(1/\epsilon))$ gates, c varies from 1-4 depending on assumptions.

Universal gate sets

Using a different gate set is like *speaking a different language*.



I would like to use a real
quantum computer.

*I would like to use a real
quantum computer.*



J'aimerais utiliser un vrai
ordinateur quantique.

*I would like to use a real
computer that is quantum.*



本当の量子コンピュータを作りたい。

*A real quantum computer, I
would like to use.*

Universal gate sets

Any single-qubit unitary can be expressed using a sequence of 3 Pauli rotations around a minimum of 2 unique axes

- ZYZ
- XYX
- XYZ
- 9 other combinations (*Euler angles*)

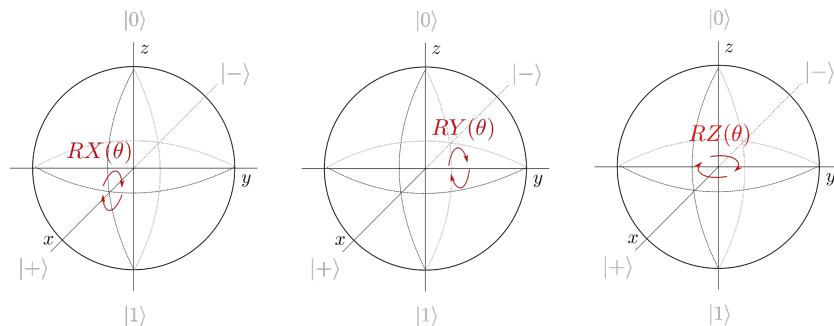
qml.Rot

```
class Rot(phi, theta, omega, wires, id=None)
```

Bases: `pennylane.operation.Operation`

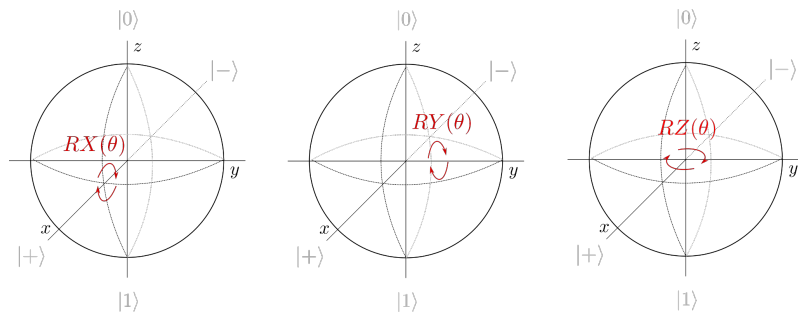
Arbitrary single qubit rotation

$$R(\phi, \theta, \omega) = RZ(\omega)RY(\theta)RZ(\phi) = \begin{bmatrix} e^{-i(\phi+\omega)/2} \cos(\theta/2) & -e^{i(\phi-\omega)/2} \sin(\theta/2) \\ e^{-i(\phi-\omega)/2} \sin(\theta/2) & e^{i(\phi+\omega)/2} \cos(\theta/2) \end{bmatrix}.$$

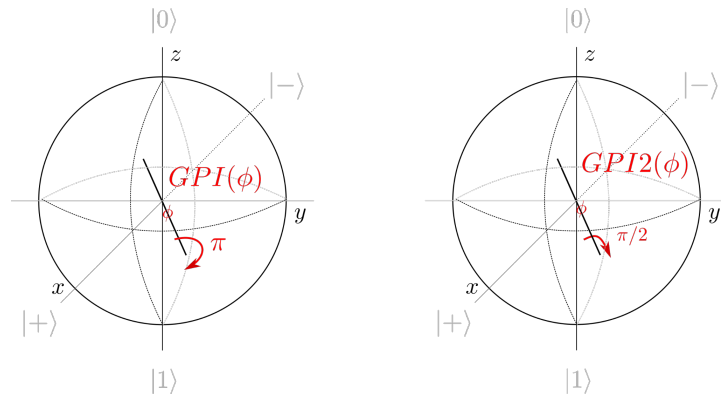


Hardware-specific example

$RX/RY/RZ$ are **arbitrary-angle** rotations about **fixed axes**

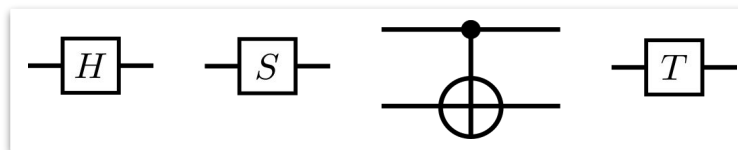


IonQ's trapped ion hardware native gates: $GPI/GPI2$, **fixed-angle** rotations about **arbitrary axes** in xy -plane

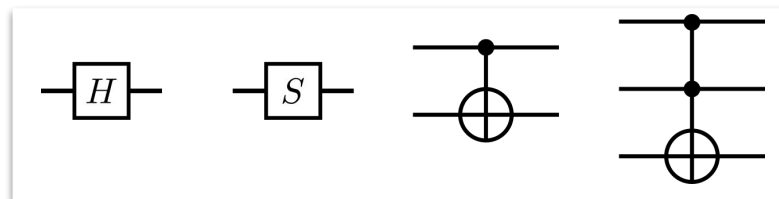
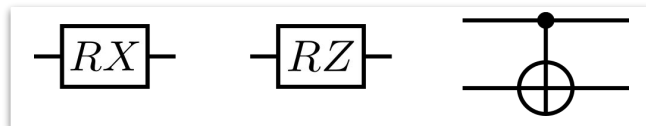


Universal gate sets

Multi-qubit gate sets

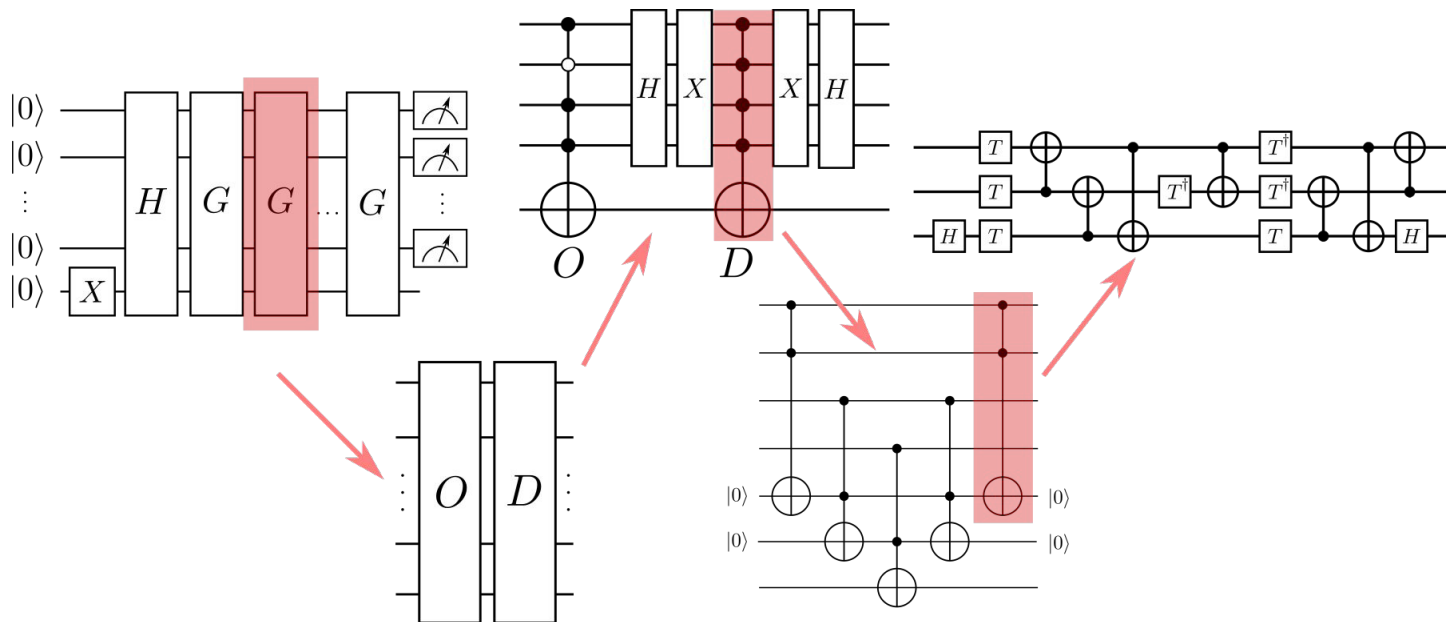


Clifford + T



Circuit decomposition

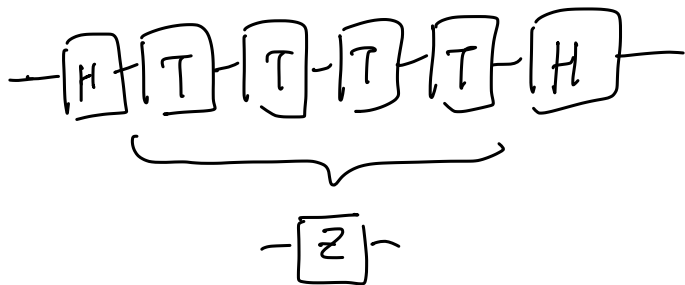
Decompositions of many common operations are **known**. Some are even straightforward to work out by hand.



Exercise

Express Pauli X in terms of H and T .

Hint: start by expressing it using H and Z .



Exercise

Express Hadamard in terms of $\{RX, RY, RZ\}$

Hint: there are many ways to do this!

$RX/RY \pi/2$

$$RY(\pi/2) = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

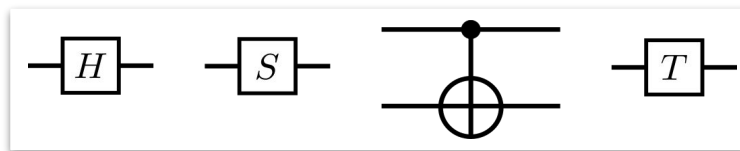
$$Z: \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \quad \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

find ways to do!
(we will use this
on Thursday)

Exercise

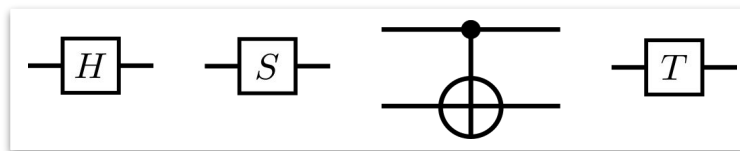
Express CZ using gates from

Try at home



Take-home exercise

Express controlled Hadamard using gates from



Try at home!

Next time

Content:

- The quantum compilation stack (from top to bottom)
- Quantum circuit optimization
- Hands-on: compilation and resource estimation of Grover's algorithm

Action items:

- Project group and paper selection due Thursday

Recommended reading:

- Nielsen and Chuang 4.5
- [Compiling quantum circuits](#) (PennyLane documentation)
- Explore the [qml.transforms](#) module